

Image Inpainting

Week 1 Project Scope and Experimental Plan

Ivan | Alexander | Egor | Tatiana | October 5, 2026

Abstract

This project develops an image inpainting application that removes masked objects or arbitrary regions and reconstructs the missing content. We will compare four approaches representing different levels of image restoration: OpenCV Telea as a classical baseline, LaMa as a CNN-based method, ViTInpainter as a Transformer-based method, and FLUX.1-Fill as a diffusion-based generative model. A controlled subset of MS COCO 2017 will be used to generate masked inputs with several shapes and sizes. Pretrained models will be evaluated using PSNR, SSIM, LPIPS, and inference time, with the main reconstruction scores computed inside the masked region. The experiment will additionally document typical semantic failures and study how mask geometry and size affect reconstruction quality.

1 Project Scope

The project objective is to build an application that removes masked objects or arbitrary regions from photographs and fills the missing area with a plausible reconstruction.

The central research question is:

How do classical, CNN-based, Transformer-based, and diffusion-based methods compare in image inpainting quality and inference speed under different mask sizes and shapes?

The exact experimental setup contains four approaches:

- **OpenCV Telea** - classical inpainting baseline;
- **LaMa** - CNN-based image inpainting;
- **ViTInpainter** - Transformer-based image inpainting;
- **FLUX.1-Fill** - diffusion-based generative inpainting.

The task is generic image inpainting rather than a single object class. Person removal will be one representative application scenario, while the main benchmark will cover arbitrary masked regions and object-level masks.

Controlled comparison

All four approaches will receive the same image, mask, image resolution, and evaluation protocol whenever the model supports the same input setting. The main experimental variables are the inpainting method and the mask condition.

2 Dataset Preparation

Dataset: MS COCO 2017.

A manageable, controlled subset of COCO will be used rather than the full dataset. COCO provides diverse real-world images and object segmentation annotations, which makes it possible to evaluate both synthetic region masking and realistic object removal.

The preprocessing pipeline will create masked inputs from original images. The original image remains the ground-truth reconstruction target. We will evaluate four main mask types:

- rectangular masks;
- circular or elliptical masks;
- free-form brush masks;
- object segmentation masks derived from COCO annotations.

For the main sensitivity experiment, synthetic masks will use three approximate area levels: **small** (5%), **medium** (15%), and **large** (30%). Object masks will be evaluated separately because their geometry is determined by the annotated object.

Split	Images	Purpose
Train / development	1,000	Pipeline development, preprocessing, and mask-generation experiments
Validation	200	Parameter selection and sensitivity experiments
Test	500	Final held-out evaluation

The selected models will be used with pretrained weights. No end-to-end training from scratch is planned. The split is therefore used for benchmark development, configuration selection, and final evaluation rather than for conventional model training.

3 Experimental Plan

The experiment compares methods across the same masked inputs. The main quality metrics are computed primarily inside the masked region so that unchanged pixels outside the mask do not dominate the evaluation.

3.1 Models and strategy

All models will initially be evaluated using their pretrained weights. No model-specific fine-tuning is required for the first experimental version. The shared benchmark will keep preprocessing, image resolution, mask conditions, and evaluation code as consistent as possible.

3.2 Evaluation metrics

- **PSNR** $\uparrow$ - pixel-level reconstruction similarity;
- **SSIM** $\uparrow$ - structural similarity;

- **LPIPS** ↓ - perceptual similarity;
- **Inference time** ↓ - computational efficiency under the same hardware and resolution.

The qualitative part of the evaluation will inspect visual naturalness, texture continuation, structural consistency, residual object traces, and semantically incorrect or hallucinated content.

Method	PSNR ↑	SSIM ↑	LPIPS ↓	Time ↓
OpenCV Telea	—	—	—	—
LaMa	—	—	—	—
ViTInpainter	—	—	—	—
FLUX.1-Fill	—	—	—	—

Table 1: Planned main comparison table. Values will be filled after the benchmark.

3.3 Sensitivity and failure analysis

The primary sensitivity experiment will study the effect of mask shape and size on reconstruction quality. We will additionally document representative semantic failures, including large missing regions, complex structures hidden behind the mask, repetitive textures, and plausible but incorrect generated content.

4 Implementation Plan

The complete pipeline will be organized around a shared inpainting interface:

```
image + mask → inpaint(image, mask) → reconstructed image
```

The main implementation steps are:

1. prepare the COCO subset and generate controlled masks;
2. implement the common preprocessing and model interface;
3. integrate OpenCV Telea and LaMa;
4. integrate ViTInpainter and FLUX.1-Fill;
5. implement PSNR, SSIM, LPIPS, and runtime measurement;
6. run the mask-size and mask-shape benchmark;
7. collect representative visual results and semantic failures;
8. expose the four inference methods through the application backend.

The target development environment is a common **Python 3.12** environment. Pretrained model checkpoints and the corresponding modern PyTorch / Transformers / Diffusers ecosystem will be used where applicable. OpenCV will provide the classical baseline.

5 Team Responsibilities

Member	Role	Responsibilities
Ivan	Team Lead / LaMa	LaMa integration, OpenCV baseline, common benchmark and evaluation pipeline, dataset/mask pipeline, experiment coordination, final integration and report coordination.
Alexander	Transformer / ViTInpainter	ViTInpainter integration, preprocessing and postprocessing, Transformer-specific benchmarking, runtime and memory measurements.
Egor	Diffusion / FLUX.1-Fill	FLUX.1-Fill integration, inference configuration, VRAM/runtime investigation, representative results and semantic failure analysis.
Tatiana	Frontend / Back-end	Application interface, image and mask input flow, back-end/API integration, result gallery, and application run configuration.

All team members will understand the shared benchmark and experimental design in addition to their individual implementation areas, since the project includes individual technical questions about model choices, metrics, results, and failure cases.